Power Master

Dynamic Load Balancing for EV Charging Infrastructure

Intelligent power distribution for multi-charger deployments

TECHNICAL WHITEPAPER

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Executive Summary

As EV adoption accelerates, multi-charger deployments face a critical challenge: limited electrical capacity. Installing multiple EV chargers without proper power management can overload circuits, trip breakers, and require expensive electrical upgrades.

Power Master is RIOD's Dynamic Load Balancing (DLB) solution that intelligently distributes available power across multiple chargers in real-time, maximising charging throughput without exceeding the site's electrical capacity.

The Problem: Electrical Capacity Constraints

Consider a typical scenario: An office building has 100A of spare capacity on its distribution board. At 230V single-phase, this provides approximately 23 kW of available power for EV charging.

Without Load Balancing

- 3 chargers at 7.4 kW each = 22.2 kW (uses 96% of capacity, no headroom)
- 4 chargers at 7.4 kW each = 29.6 kW (exceeds capacity, trips main breaker)

With Power Master DLB

- 6 chargers at up to 7.4 kW each, dynamically sharing 23 kW
- Power allocated based on real-time demand, priority, and charging state
- More chargers installed on same electrical infrastructure

How Power Master Works

Power Master is a hardware device installed at the distribution board level. It continuously monitors total power consumption and dynamically adjusts charging current allocated to each POWERPOD charger.

Architecture

- Power Master hardware installed at the Distribution Board
- Connected to POWERPOD chargers via communication bus
- Real-time power monitoring using CT (Current Transformer) clamps
- Dynamic current allocation algorithm running locally on the device
- No cloud dependency for load balancing decisions

Algorithm Logic

1. Measure total site power consumption in real-time
2. Calculate available power headroom (total capacity minus current consumption)
3. Distribute available power across active chargers based on priority rules
4. Continuously adjust as vehicles connect, disconnect, or change state
5. Never exceed the configured maximum site capacity

Communication Protocols

Supported Protocols	
Zigbee	Wireless mesh for distributed parking areas
BLE	Short-range wireless for nearby chargers
RS485	Wired serial for reliable industrial deployments
Wi-Fi	Standard wireless for existing networks
Modbus	Industrial protocol for BMS integration

Modular Application Architecture

Power Master uses a modular software architecture where each function is an independent module. Operators enable only the features they need.

- Core DLB Module: Real-time power distribution across chargers
- Priority Management: Assign charging priority to specific chargers or users
- Scheduling Module: Time-based power allocation (e.g., off-peak hours)
- Solar Integration Module: Prioritise solar energy for EV charging
- Building Integration Module: Coordinate with HVAC and lighting loads
- Reporting Module: Detailed energy distribution reports and analytics
- API Module: REST API for third-party BMS integration

Deployment Scenarios

Scenario 1: Office Building (10 Chargers, 50A Available)

10x POWERPOD Go 7.4 kW installed. Without DLB: Only 6 chargers operate simultaneously. With Power Master: All 10 active, power dynamically shared (2.3 kW to 7.4 kW each based on demand).

Scenario 2: Apartment Complex (20 Chargers, 63A Available)

20x POWERPOD Lite 3.3 kW across resident parking. Without DLB: Only 19 chargers operate. With Power Master: All 20 active with headroom. Overnight charging optimised for off-peak rates.

Scenario 3: Commercial Parking (4 City Chargers, 100A Three-Phase)

4x POWERPOD City 22 kW (8 charging points). Without DLB: Only 3 units at full power. With Power Master: All 4 active, power balanced across phases. Up to 8 vehicles charging simultaneously.

Technical Specifications

Power Master Specifications	
Input Voltage	230V AC (single phase) or 415V AC (three phase)
CT Rating	100A / 200A / 400A (selectable)
Max Chargers	Up to 32 per Power Master unit
Communication	Zigbee, BLE, RS485, Wi-Fi, Modbus
Response Time	Less than 500ms for load adjustment
Operating Temperature	-20 to 60 degrees C
Mounting	DIN rail (distribution board)
Power Consumption	Less than 5W
Enclosure	IP20 (indoor DB installation)

Benefits Summary

- Install more chargers on existing electrical infrastructure
- No expensive electrical upgrades or transformer changes
- Real-time, local load balancing with no cloud dependency
- Supports mixed charger configurations (different power ratings)
- Modular features - enable only what you need
- Multi-protocol support for flexible deployment
- Future-ready architecture for solar integration and building coordination

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